Chapter

Virtual Instrumentation Used in Renewable Energy

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Abstract

The demand of energy increases once with the growth of population, and therefore, the finding of or improvement in the efficiency of renewable energy sources becomes very important for researchers and industry. The conversion of solar energy into electrical energy can be done based on photovoltaic or Seebeck effects. In the first case, photovoltaic panels are used, while in the second case, thermoelectric generators are used. The two can be combined to obtain the so-called hybrid systems, which have the goal to improve the overall conversion efficiency of the system. This chapter is focused on showing how the graphical programing language, called NI LabVIEW, together with a SPICE simulator, called NI Multisim, can be used for studying and understanding the behavior of the photovoltaic and thermoelectric generators as parts of the renewable energy sources. Different simulations developed in LabVIEW or Multisim are presented, and some monitoring and characterization applications are also described. Simple simulations to complex laboratory or industrial-level applications are dealt with in this chapter.

Keywords: virtual instrumentation, LabVIEW, simulations, real-time and FPGA platforms, photovoltaic cells, thermoelectric generator

1. Introduction

According to a recent analysis [1], the world's population reached 8 billion people in November 2022. Therefore, the energy demand of the entire population is increasing rapidly. The actual used energy is obtained from different energy sources (ES), but unfortunately, the largest percent of these ES is based on fossil resources, which are very polluting and limited. In order to have a sustainable civilization from the energy point of view, renewable energy sources (RES) should be used. RES include solar energy, wind energy, geothermal energy, biomass-based energy, waves energy, and so on.

One problem of using RES is the variability of energy availability in time. For example, in the solar energy case, the availability depends on the moment of time – day or night – and the climate conditions – cloudy or clear sky. Therefore, the usage of this RES must be combined with another or implemented with energy storage solutions – like batteries. Such solutions are called hybrid solutions. The solar

energy is converted into electrical energy and thermal energy. In the first case, the most used conversion technologies are based on the photovoltaic and Seebeck effects. The photovoltaic effect is exploited using photovoltaic cells and panels (PV). The Seebeck effect is exploited using thermoelectric generators (TEG). The conversion of solar energy into thermal energy is done by using so-called solar thermal collectors, which absorb the solar radiation and heat the internal thermal agent, which can be water, air, nanofluid, or oil [2]. The combination of these energy conversion systems is used nowadays more often in order to improve the overall efficiency of the system. There are different hybrid systems for solar energy conversion, such as [3]:

- Photovoltaic-thermal (PV-T) system that uses the PV to convert the solar irradiance into electricity and solar collector to heat the thermal agent but also to cool down the PV in order to increase the PV efficiency and its life time.

- Photovoltaic-thermoelectric generator (PV-TEG) system that uses both components to generate electricity. The PV absorb around 95% of the solar irradiance and convert, depending on the PV type, between 10 and 25% into electricity, while the rest is converted into heat, which increases its temperature and decreases its efficiency. The TEG based on the Seebeck effect absorbs the heat from the PV and generates electricity.

- Photovoltaic-thermoelectric generator-thermal (PV-TEG-T) system that combines all three components to generate electrical and thermal energy.

In order to study or monitor RES-based systems, different software and hardware should be used. There are many solutions in the market; some of them are dedicated to a specific RES, while others are dedicated to the general use.

This chapter is focused on using the NI LabVIEW graphical programming languages for RES study. The NI LabVIEW is used at all levels: simulation level, measurement and characterization level, and monitoring and industrial level.

NI LabVIEW is a development software platform that allows the implementation of the virtual instrumentation (VI) concept. According to NI, VI represents "a combination of modular hardware and customizable software used to create measurement and control system defined by user." In fact, a virtual instrument is a measurement and control instrument based on a computing system (in the majority of cases, it is based on a PC). The VI is an alternative to the traditional instrumentation (TI). The TI normally has a functionality and user interface defined by the manufacturer. Instead, the VI functionality and user interface are defined by the user and can be easily updated or replaced.

Considering the advantages offered by the VI and NI LabVIEW, these are used in this chapter to develop simulated and real instruments for RES study, focused on PV systems and some hybridization solutions.

The chapter is split into four sections, as follows: in the second section is discussed the implementation of the simulations for PV and TEG components based on different tools. The third section is dedicated to the implementation from simple to complex methods for PV and TEG characterization. The fourth section shows how a monitoring system based on a real-time (RT) platform for a home PV system is implemented. This section will show how to use VI in Industrial Internet of Things (IIoT) implementation.

2. Simulation

This section is dedicated to show how to use the NI LabVIEW for PV and TEG study through simulations.

2.1 Theoretical aspects: PV

The PV are used for converting the solar irradiance to electricity. A PV panel is composed of a set of PV cells combined in a series and/or parallel connection in order to obtain the desired voltage and current. The most used model of a PV cell is the so-called one diode model, which is shown in **Figure 1**.

The diagonal cross fill from **Figure 1** marks the one diode model of the PV whose mathematical description is given by eq. (1).

$$I = I_{ph} - I_0\left(e^{\frac{q(V+IR_s)}{nk_BT}} - 1\right) - \frac{V + IR_s}{R_{sh}} \tag{1}$$

where V and I are the output voltage and current of the PV cell, I_{ph} represents the photogenerated current, I_0 is the reverse saturation current, q is the elementary electrical charge, R_S is the series resistance of the PV cell, R_{SH} is the shunt resistances of PV cell, n is the ideality factor of the diode D, k_B is the Boltzmann constant, and T is the PV cell temperature. The quantity V_T (2) is called thermal voltage of the diode and has a value ~26 of mV at room temperature.

$$V_T = \frac{k_BT}{q} \tag{2}$$

By connecting N_S PV cells in a series, a PV panel can be obtained. The PV panel's mathematical description is given by eq. (3).

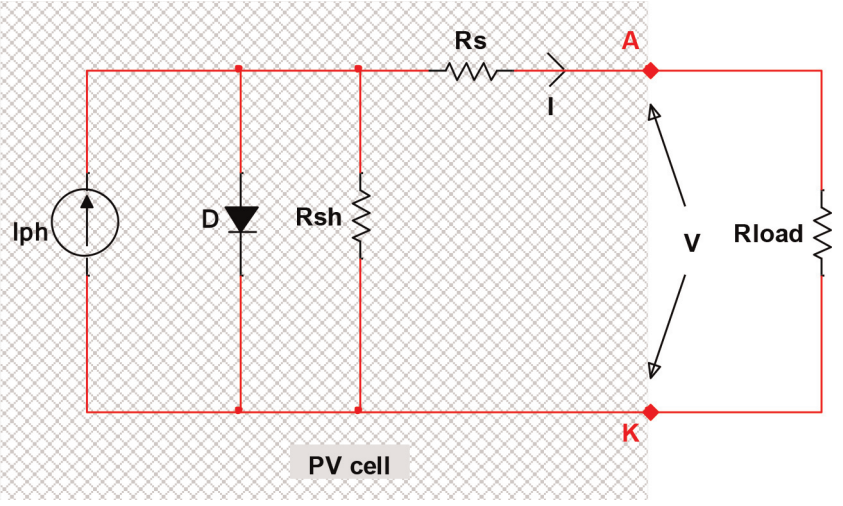


Figure 1.
The one diode model of a PV cell.

$$I = I_{ph} - I_0\left(e^{\frac{q(V+N_sIR_s)}{N_S n k_B T}} - 1\right) - \frac{V + N_s I R_s}{R_{sh}} \tag{3}$$

where all parameters refer now to a PV panel and not to a PV cell.

Considering the PV panel parameters offered by the manufacturer, such as: V_{OC} – open circuit voltage, I_{SC} – short circuit current, k_V and k_I – current and voltage thermal coefficients of the PV panel, R_{SH}, R_S, n, and N_S, the current-voltage (I-V) characteristics of the PV panel can be determined for different levels of irradiance and temperature using the following equations:

$$V_{OC}(T) = V_{OC}(T_{REF}) + k_V(T - T_{REF}) \tag{4}$$

$$I_{SC}(T) = I_{SC}(T_{REF})\left(1 + \frac{k_V}{100}(T - T_{REF})\right) \tag{5}$$

$$I_0(T) = \left(I_{SC}(T) - \frac{V_{OC}(T) - I_{SC}(T)R_S}{R_{SH}}\right)e^{-\frac{V_{OC}(T)}{N_S V_T}} \tag{6}$$

$$I_{PH}(T) = I_0(T)e^{\frac{V_{OC}(T)}{N_S V_T}} + \frac{V_{OC}(T)}{R_{SH}} \tag{7}$$

$$I_{SC}(G) = I_{SC}(T)\frac{G}{1000} \tag{8}$$

$$I_{PH}(G) = I_{PH}(T)\frac{G}{1000} \tag{9}$$

$$V_{OC}(G) = ln\left(\frac{I_{PH}(G)R_{SH} - V_{OC}(T)}{I_0(T)R_{SH}}\right)N_S V_T \tag{10}$$

$$I(G,T) = I_{PH}(G) - I_0(T)\left(e^{\frac{(V+I(G,T)R_S)}{N_S V_T}} - 1\right) - \frac{V + I(G,T)R_S}{R_{SH}} \tag{11}$$

$$D_V = V_{OCi}(G) - V_{OCi-1}(G) \tag{12}$$

$$D_I = I_i(G,T) - I_{i-1}(G,T) \tag{13}$$

where T_{REF} is the reference temperature (25°C), and G is the irradiance level. Eqs. (10) and (11) are implicit ones and therefore are calculated iteratively. The stop conditions for the iterations are achieved by the predefined maximum number of iterations or the difference between values obtained at two consecutive iterations D_V and D_I. The calculus of these differences is based on eqs. (12) and (13), respectively.

2.2 Theoretical aspects: TEG

A TEG is a device that allows the conversion of thermal energy into electrical energy. It is based on the Seebeck effect, which consists of an electromotive force that appears at the terminals of an electrical circuit made up of two different types of materials, when there is a temperature difference between the terminals. A TEG consists of a set of thermo-elements (modules or uni-couples) connected in series. A thermo-element is formed by two legs: a p-type and an n-type semiconductor, connected in series through a metallic conductor (copper), as shown in **Figure 2a**. The simple equivalent electric circuit of a TEG consists of a thermal dependent voltage source connected in series with a resistor (**Figure 2b**).

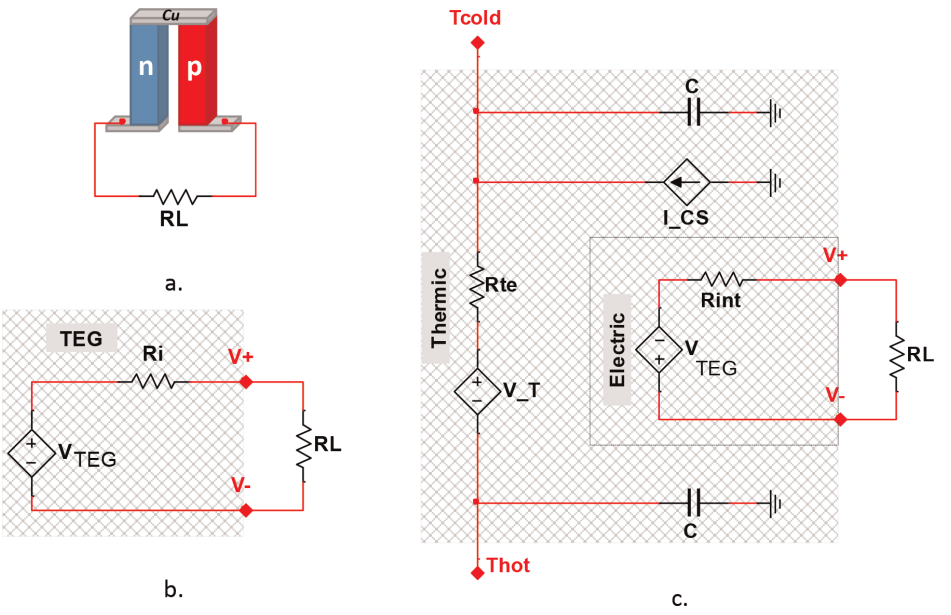


Figure 2.
The model of a TEG. a. the TEG pellet; b. simple model of a TEG; and c. thermal and electrical model of a TEG.

In order to model the thermal dependence of the voltage source, the equivalent circuit must be updated as shown in **Figure 2c**, which is described by the following equation:

$$V_T = IR_{te}\left(\alpha_{Sb}T_{hot} - \frac{IR_{int}}{2}\right) \tag{14}$$

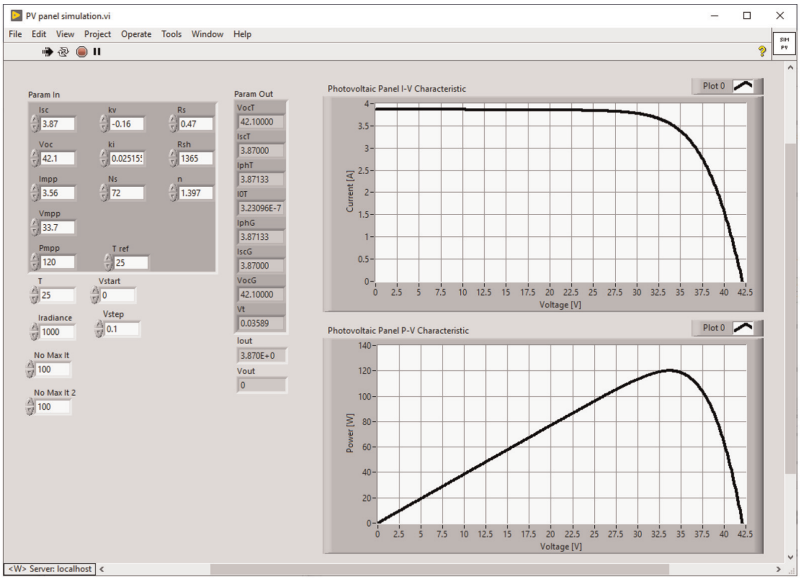
$$V_{TEG} = \alpha_{Sb}(T_{hot} - T_{cold}) \tag{15}$$

$$I_CS = V \cdot I \tag{16}$$

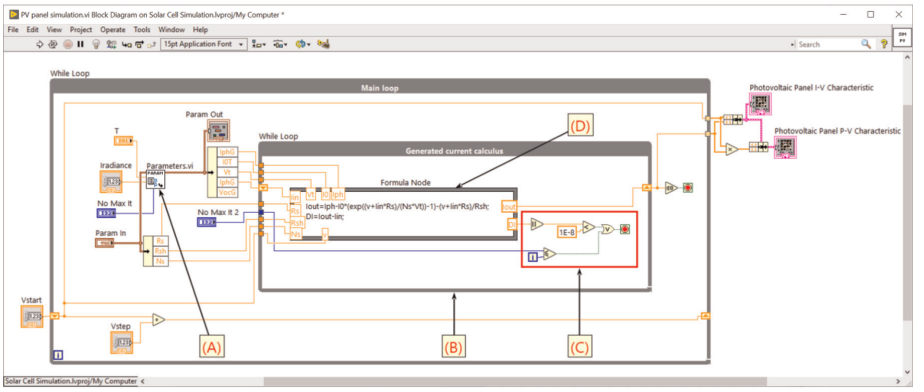
where V_T is a voltage source that models the Peltier cooling and heating on the cold and hot sides, R_{te} is the thermal resistance of the TEG, α_{Sb} is the Seebeck coefficient, R_{int} is the electrical internal resistance of the TEG, V_{TEG} is the voltage source that corresponds to the Seebeck effect, I_CS is the current source that corresponds to the Joule heating of the TEG, I is the output current generated by the TEG, V is the output voltage of the TEG, T_{hot} and T_{cold} are the temperatures of the hot and cold sides of the TEG, *and* C is a capacitor that models the lumped heat capacitance of the ceramic plates of the TEG.

2.3 LabVIEW simulation

A possible LabVIEW implementation of the PV panel simulation is shown in **Figure 3**. **Figure 3a** represents the user interface of the main LabVIEW application that allows users to introduce the PV panel parameters, **Param In**; the level of irradiance, **G**; and the PV working temperature, **T**. The user can also introduce the starting voltage and voltage step for the I-V characteristics calculus. On the two graphs of the



a.

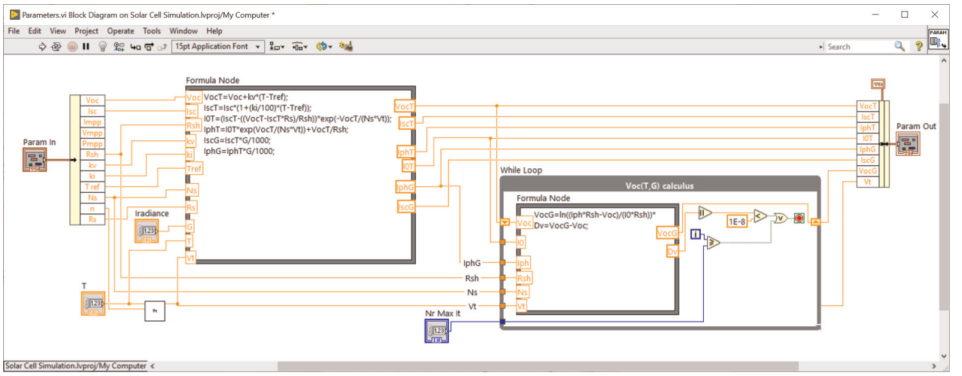


b.

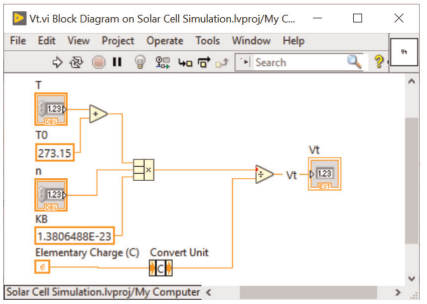
Figure 3.
LabVIEW implementation of the PV panel simulation.

application are shown the I-V and power-voltage (P-V) characteristics of the PV panel. The application also displays the PV panel parameters for the set temperature and irradiance level, **Param Out**.

Figure 3b presents the LabVIEW code for PV panel simulation. The Parameters.vi (A) is a subVI used to determine the PV parameters at the desired condition (T and G) based on eqs. (4)–(10) and (2) (see **Figure 4a** and **b**). The generated current, I, is calculated using the *Formula Node* structure (D). The current is determined iteratively using the *While Loop* structure (B), which uses D_I as finish conditions whose value should be smaller than 10^{-8} or the maximum number of iteration (C). The same implementation approach is used to determine the value of V_{OC} for the desired T and G (**Figure 4a**). In **Figure 4b**, the value of V_T is determined using eq. (2).



a.



b.

Figure 4.
PV panel parameters calculus.

2.4 LabVIEW and NI Multism simulation

2.4.1 PV simulation

An interesting option offered by LabVIEW is to use the NI LabVIEW Multisim API Toolkit that allows to interconnect the LabVIEW programming languages with a SPICE simulator. NI Multisim is an application for SPICE simulation and circuit design [4] and together NI Ultiboard represents a complete software package for circuit design, simulation, validation, and PCB layout.

If the LabVIEW Multisim API Toolkit is installed, the corresponding library is available in the function palette of LabVIEW (**Figure 5**) at the following path *Function> > Connectivity> > Multisim*.

Developed models of PV cells and PV panels in NI Multisim using SPICE code are described in the study case "New Models for Photovoltaic Cells in Multisim" published on the NI website (see [5]). The PV panel model developed is based on eqs. (2)–(11) (**Figure 6**). The PV model is encapsulated in the **U1 PV** component.

Based on the *Parameter Sweep* analysis, the I-V characteristic can be easily obtained in Multisim. The used parameters of the analyses are shown in **Figure 7a**. The output parameter is set in the *Output* tab as the current through the V_{bias} source. This source is a voltage source for PV panel polarization. V_{irrad} is used to fix the level of irradiance. A value of 1000 V means an irradiance level of 1000 W/m^2 (1 sun). **Figure 7b** shows the obtained I-V characteristics of the PV panel.

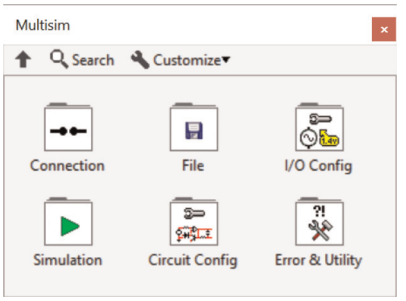


Figure 5.
The LabVIEW Multisim palette.

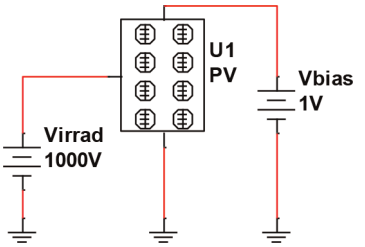
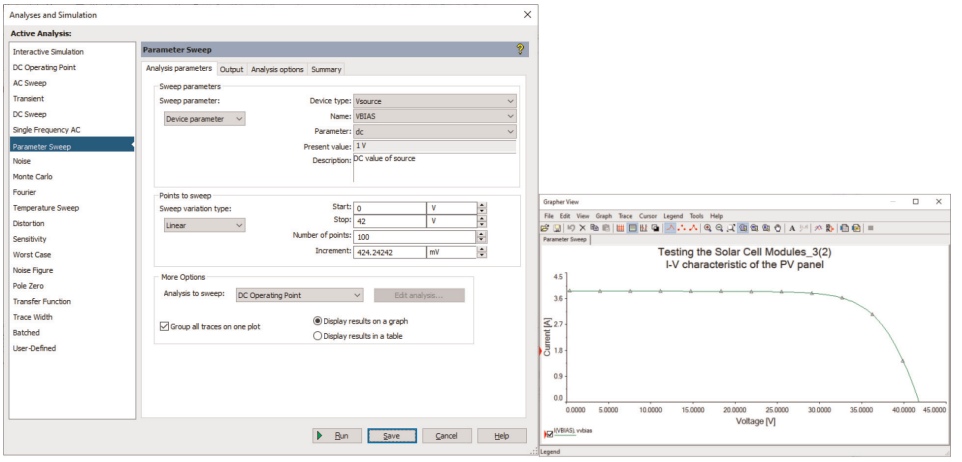


Figure 6.
The Multisim model of a PV panel.



a. b.

Figure 7.
The Multisim model of a PV panel.

The Multisim allows combining the variation of more parameters during the same analyses, so that the effect of the irradiance or temperature variation could be easily studied. In **Figure 8**, the Parameter Seep setup for studying the influence of the irradiance level over the PV panel response is shown. The value of the V_{irrad} source is used as the first sweep parameter. The range of variation is fixed between 400 and 1000 with a step of 200. In the *More option* field, the *Nested sweep* option is chosen (**Figure 8a**). By pressing the *Edit analysis* button, the second sweep parameter is set

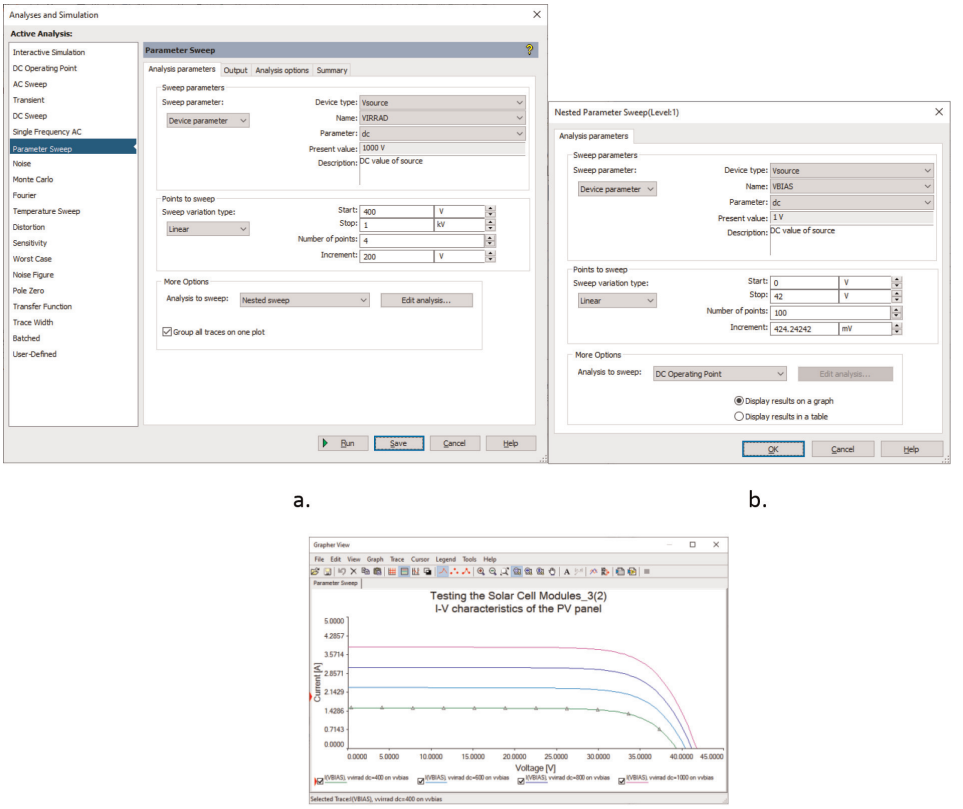


Figure 8.
The influence of the irradiance level on the PV panel response using nested sweep analysis.

(**Figure 8b**). The range value of the V_{bias} source is fixed between 0 and 42 volts with 100 equidistant number of points. The *Analysis to sweep* option is fixed to the *DC Operating Point* value. The obtained results are shown in **Figure 8** c.

These analyses can be performed directly from LabVIEW by using the NI LabVIEW Multisim API Toolkit. The advantage of such an approach is the fact that the data are obtained directly in the LabVIEW environment and can be easily compared to the data obtained from real experiments.

In order to be able to call the Multisim circuit directly from LabVIEW, it is necessary to add desired probes in the circuit (**Figure 9**). These probes become circuit outputs, which can be read in the LabVIEW application. **Figure 9** shows the PV panel circuit modified by adding two probes for current, $\boldsymbol{I_{out}}$, and voltage, $\boldsymbol{V_{out}}$, reading.

Based on Multisim API subVIs, a LabVIEW application was developed to interrogate the Multisim circuit. In **Figure 10**, the panel and diagram of the application are presented. The *Connect.vi* sets the connection between LabVIEW and Multisim; then, the circuit is opened using *Open File.vi*, which has the path of the circuit file as input. *Set Input Data.vi* and *Ramp Pattern.vi* are used to set the values of the $\boldsymbol{V_{bias}}$ voltage source. *Enum Outputs.vi* allows to find all probes defined in the circuit. In this example, all probes are read and therefore are passed to the *Set Output Request.vi* in order to have the values of the voltage and current determined through simulation directly in LabVIEW. After the circuit is configured, the simulation is started with *Run*

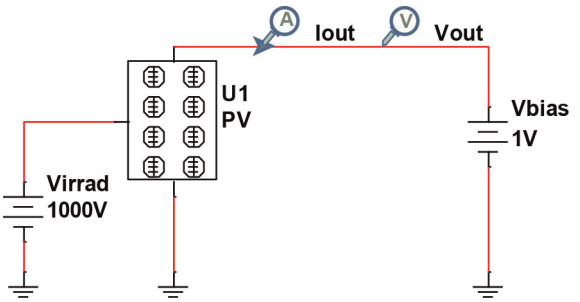


Figure 9.
The modified Multisim circuit with current and voltage probes.

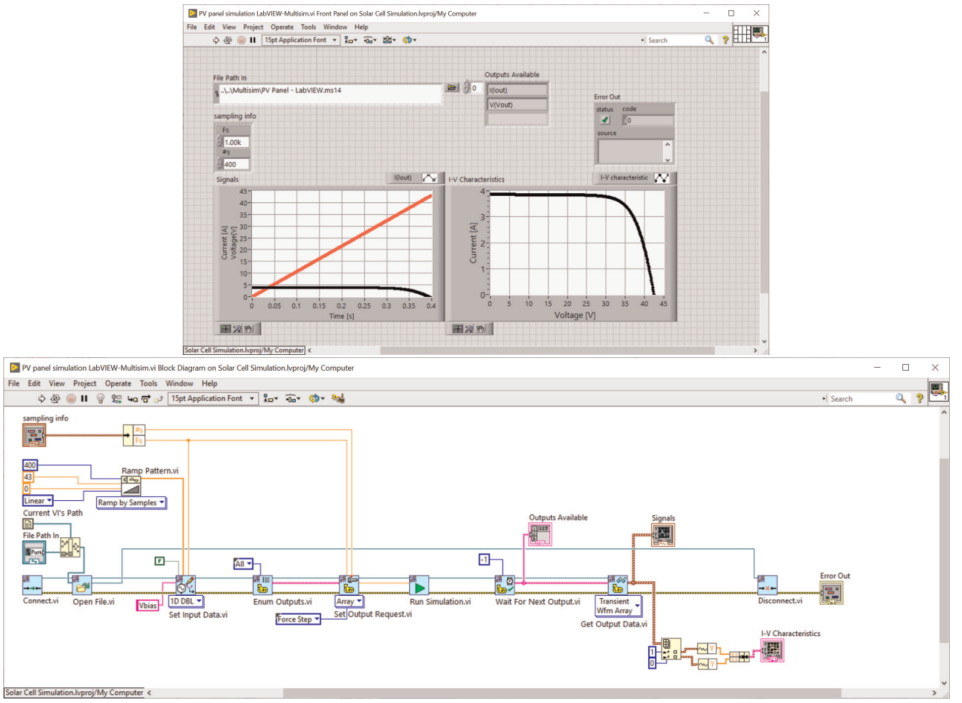


Figure 10.
The LabVIEW application developed with NI LabVIEW Multisim API.

Simulation.vi, and with *Wait for Next Output.vi*, the simulation ending is monitored. *Get Output Data.vi* returns data obtained through simulation, which are displayed on a Waveform Graph as individual signals (current and voltage generated by the PV panel) and as I-V characteristics. *Disconect.vi* is used to close the connection between LabVIEW and Multisim.

A very important parameter that considerably influences the PV panel efficiency is the temperature of the PV panel. Therefore, by adding the temperature parameter in the Multisim circuit (**Figure 11**), its influence on the PV panel efficiency can be easily investigated.

The LabVIEW application should now include the *Set Circuit Parmeters.vi,* which allows modifying the TEMP parameters and through it the PV panel behavior (see **Figure 12**). By introducing the simulation code in a *For Loop* and using the *Simulation*

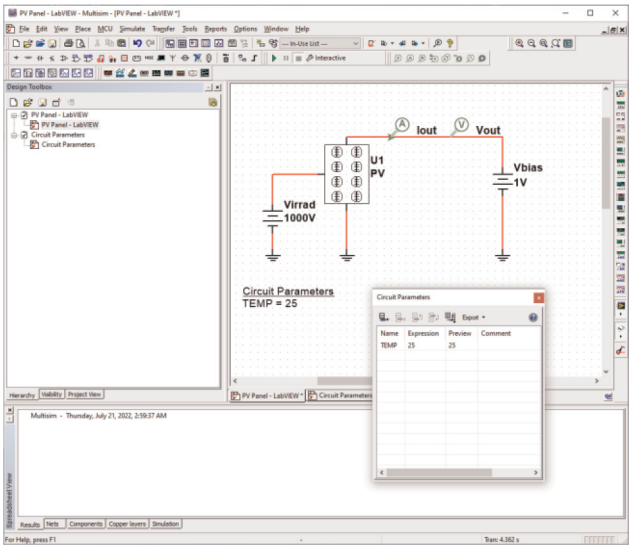


Figure 11.
Activating the TEMP circuit parameter in Multisim.

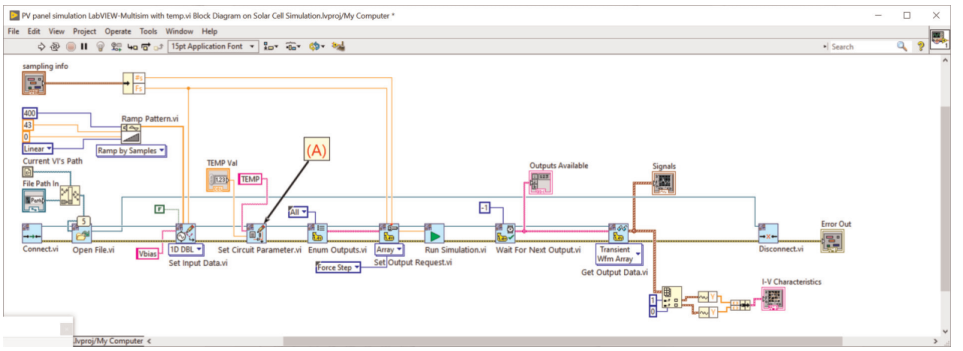


Figure 12.
The LabVIEW application with TEMP circuit parameter.

State.vi and *Simulation Stop.vi,* as shown in **Figure 13**, the effect of temperature over the PV panel response can be visualized (see the graph in **Figure 13**).

2.4.2 TEG simulation

The same approach as in the case of PV panels can be applied in the case of TEGs. The TEG model developed using the SPICE code is presented in [6] and is based on eqs. (14)–(16). The TEG model is encapsulated in the **U2 TEG** component (**Figure 14**).

The T_{hot} and T_{amb} DC sources model the temperatures of the TEG sides, cold and hot. The V_{pol} is a voltage source used for biasing the TEG. Using the Parameters Swipe analysis, the I-V characteristics of TEG at different temperatures can be obtained. The Parameters Sweep setup with the Nested Sweep option is shown in **Figure 15a** and **b**. **Figure 15c** shows the obtained I-V characteristics of the TEG, for the hot side

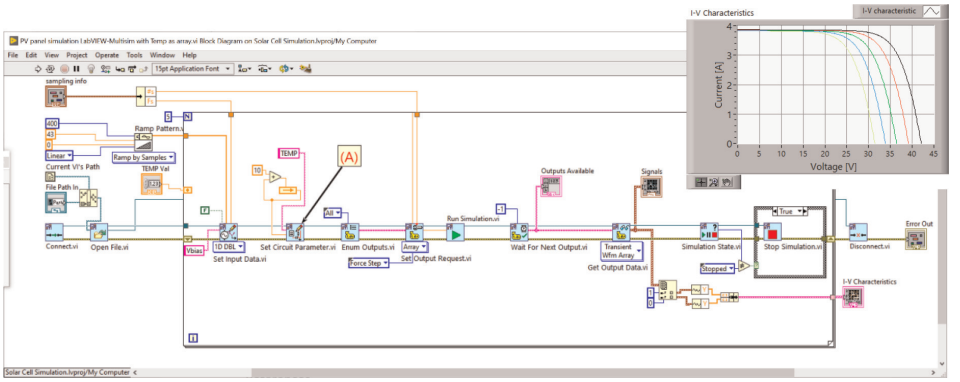


Figure 13.
The LabVIEW application with multiple values of TEMP circuit parameter.

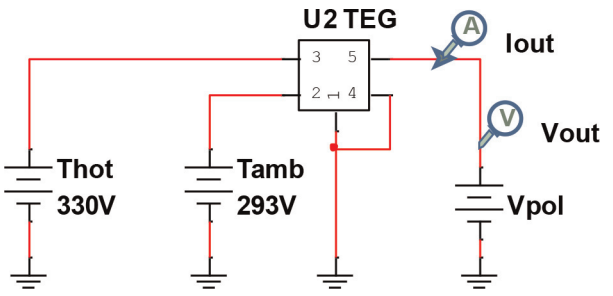


Figure 14.
The Multisim model of a TEG panel.

temperature varying between 300 and 330 K, considering that the cold side temperature is constant at 293 K.

Using the LabVIEW application shown in **Figure 10**, adapted to the new schemata, the I-V characteristic can be obtained. The modification consists of changing the name of the used source from V_{bias} to V_{pola1} and the range of the signal applied from 0 to 42 V to 0–0.4 V, as can be seen in **Figure 16**. The temperature for T_{hot} was set at a value of 330 K.

2.5 Control and simulation toolkit

Another option for implementing the simulation of PV systems is to use the LabVIEW Control Design and Simulation (CDS) toolkit. This toolkit allows to develop, implement, and analyze the behavior of different kinds of dynamic systems. In this chapter, we focus on using the simulation part of this toolkit applied in PV simulation [7]. **Figure 17** presents the simulation palette that includes libraries like Signal Generation, Signal Arithmetic, Continuous Linear Systems, Nonlinear Systems, Discrete Linear Systems, etc. One particular library is dedicated to accessing external models defined on other software and saved as shared library – dll or accessing Multisim models. Next, the use of the second option is presented. The used model is a modified model of the one presented in **Figure 6**. In this model (**Figure 18**), we used a voltage-controlled source ***V1*** as the PV panel biasing source. The ***Irad***, ***Vbias***, ***Iout,*** and ***Vout*** are hierarchical connectors (the first two are inputs and the last two are outputs).

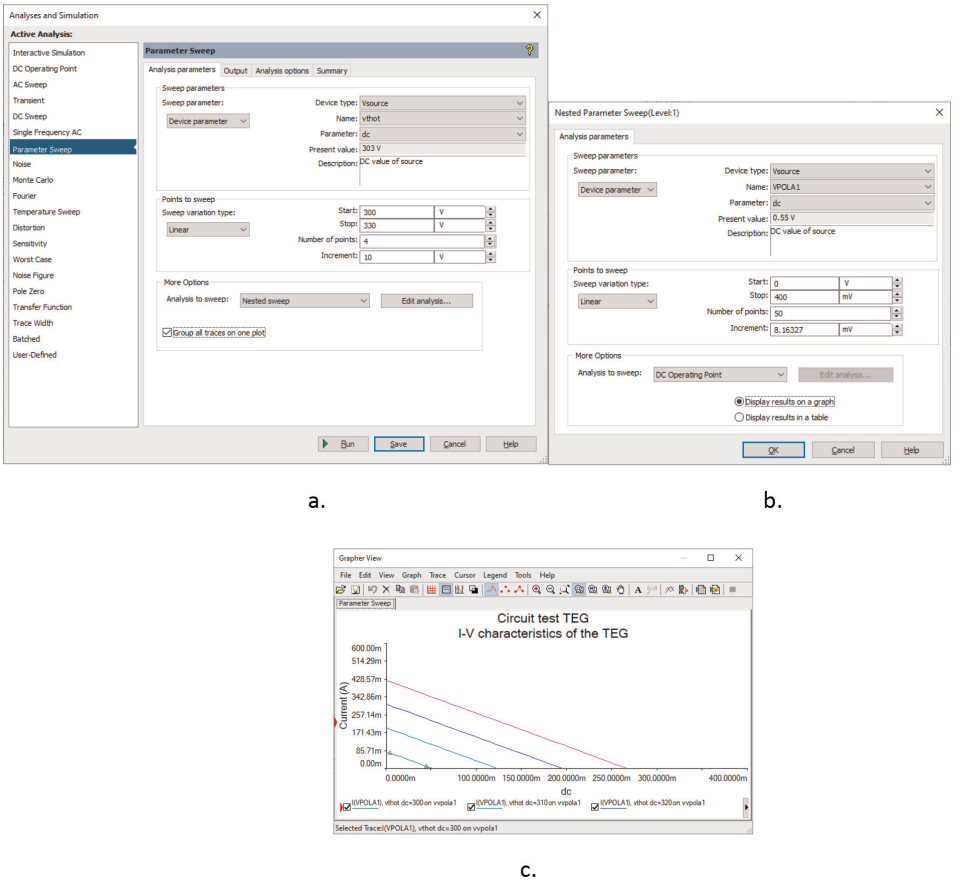


Figure 15.
The influence of temperature on the TEG response using the nested sweep analysis.

The first input is used for defining the irradiance incident on the PV, while the second one is used for controlling the ***V1*** source.

In order to read the PV current, a current clamp (XCP1) was used, with 1 mV/1 mA conversion ratio. The hierarchical connectors were used as model inputs and outputs in the LabVIEW application.

In order to develop a simulation application in LabVIEW, the *control & simulation loop* (see the red rectangle in **Figure 17**) is mandatory to be defined. The simulation functions can be used only inside of this loop.

A very simple application for PV simulation was developed in LabVIEW using the modified PV model (**Figure 18**) and the simulation functions of the CDS toolkit. The front panel and block diagram are shown in **Figure 19**. First, the *control & simulation loop* is set. Then, the *Multisim Design* node is used for Multisim model accessing. This node can be found at the following path: *Functions> > Control and Simulation> > Simulations> > External Models> > Multisim*. The node requires the path for the Multisim file. This node recognizes automatically the defined inputs and outputs in the Multisim file.

Therefore, on the ***Irrad*** input, a control is connected in order to have the possibility to modify the irradiance level incident on the PV panel. To control the PV bias, a *Ramp Signal* node (*Functions> > Control and Simulation> > Simulations> > Signal*

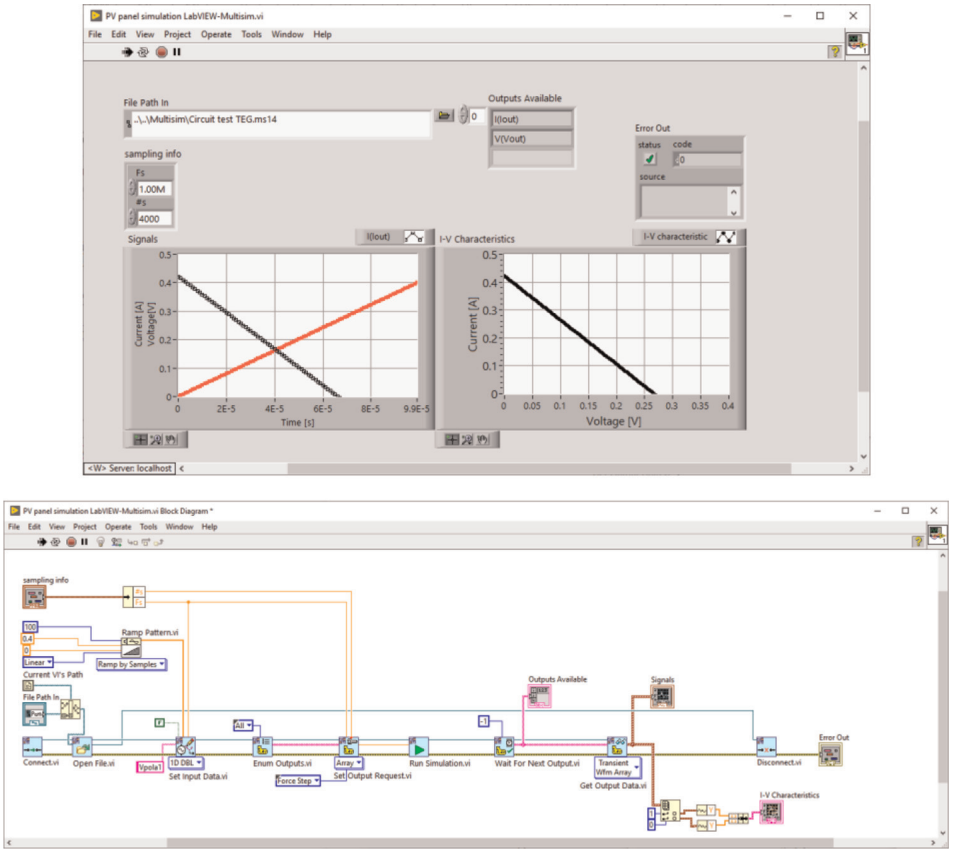


Figure 16.
The front panel and bloc diagram of the LabVIEW application for TEG.

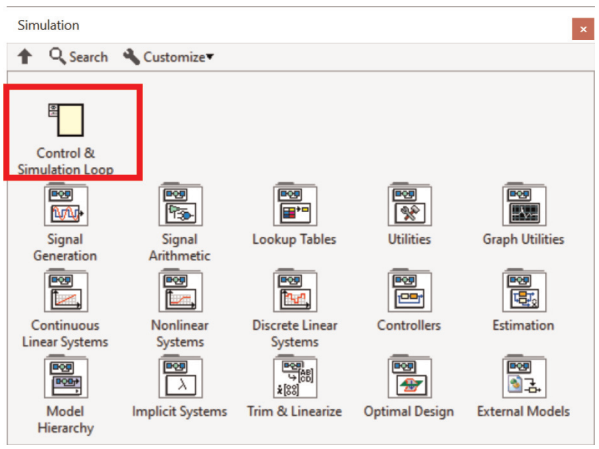


Figure 17.
The simulation palette of LabVIEW control design and simulation toolkit.

Generation) is defined and configured so that its ***Slope*** parameter is available as a terminal and starts with the initial value fixed as zero. The output of this signal generator is connected to the ***Vbias*** input. The I-V and P-V curves are built based on

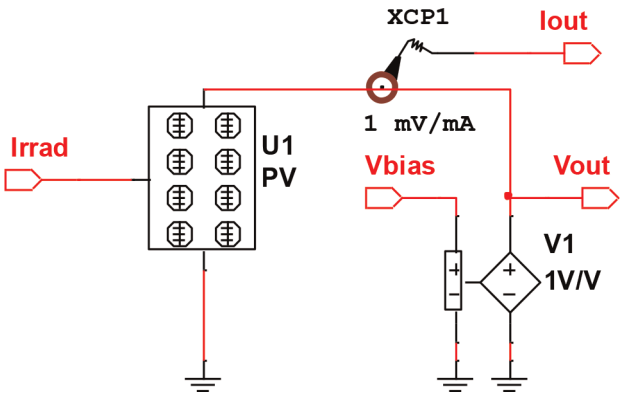


Figure 18.
The modified PV model developed in Multisim.

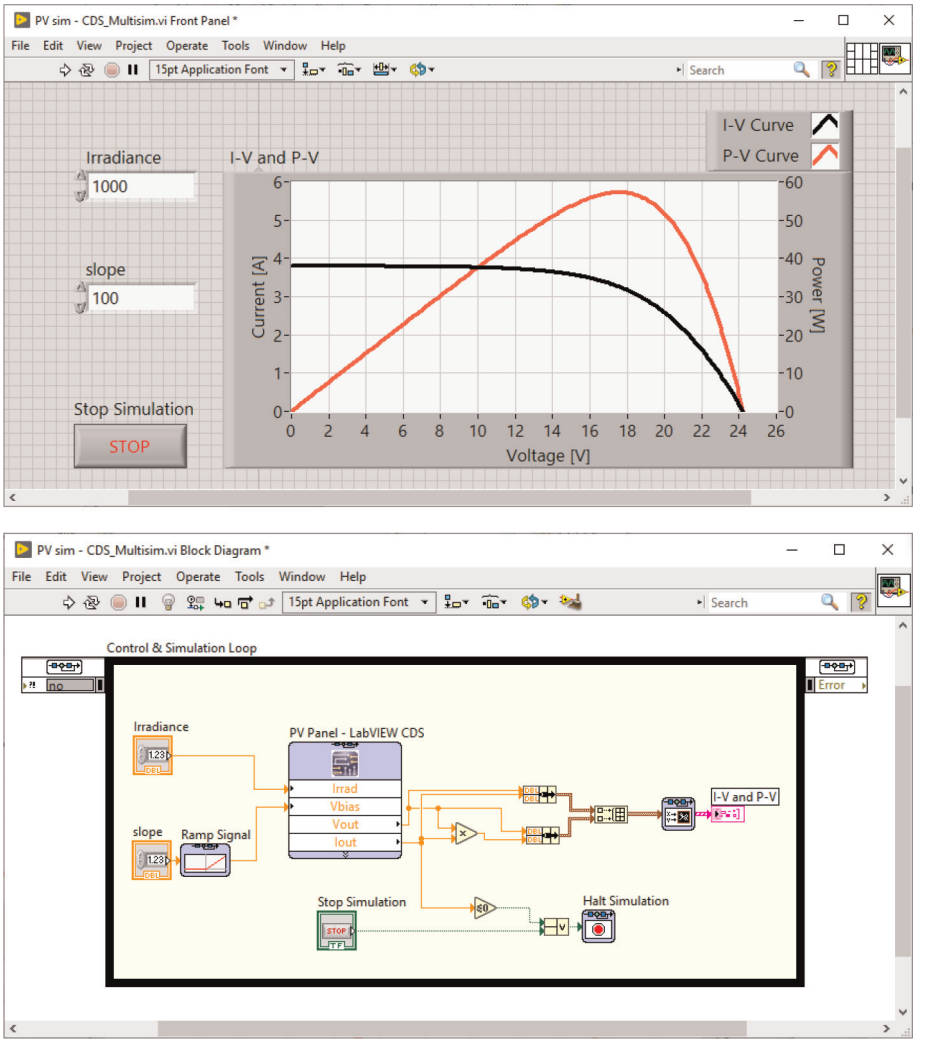


Figure 19.
The LabVIEW application developed with the CDS toolkit.

two *Bundle* functions, unified as an array with the *Build Array* function and then displayed on a *Buffered XY Graph* (*Functions*> > *Control and Simulation*> > *Simulations*> > *Graph Utilities*). The simulation is ended if the output current becomes less than or equal to zero or if the ***Stop Simulation*** button is pressed. For stopping the simulation loop, the *Halt Simulation* node (*Functions*> > *Control and Simulation*> > *Simulations*> > *Utilities*) is used.

3. Measurements and characterization

3.1 LabVIEW and simple I-V characteristics measurements

In this section, simple to complex methods for measurement of the I-V characteristics of PV panels are presented. There are many methods used for I-V characteristics measurements [8]. The simplest methods for I-V characteristics measurements are based on using dynamic electronic loads built based on a MOSFET transistor or a capacitor. The circuit schematic of dynamic loads based on the two components are shown in **Figure 20** (a shows the schematic for the MOSFET method, while b shows the capacitor method). In the MOSFET method case, by applying a ramp voltage between 0 and 5 V to the gate of the MOSFET (***Q1***), this passes form the blocked state to the completely open state. Therefore, the transistor behaves like a variable resistor, whose resistance varies from very high values to very low values. This variable resistor is applied to the PV panel, which passes from the open circuit point to the short circuit point, and by measuring the current and voltage at the PV terminals, the I-V characteristics can be determined. In the case of the capacitor method, through the SPDT switch (***S1***), the completely discharged capacitor (***C1***) is connected to the PV panel (***S1*** is on position 1). During the charging process, the internal impedance of ***C1*** varies from very low values to very high values and therefore behaves as a dynamic load for the PV panel. When the ***S1*** is passed on position 2, the ***C1*** discharges through the resistor ***R6***.

For a better measurement accuracy, the four-wire measurement method should be used, which is highlighted with thick blue circles in **Figure 20b**. For the current measurement, a power resistor with well-known resistance is used (***Rsense***). Measuring the voltage drop across the resistor and applying the Ohm law, the current value can be determined.

The resistors ***R2–R5*** are used as bias resistors for the differential setup of the DAQ device. The channels of the DAQ device used are as follows: ***AO0*** is the first analog

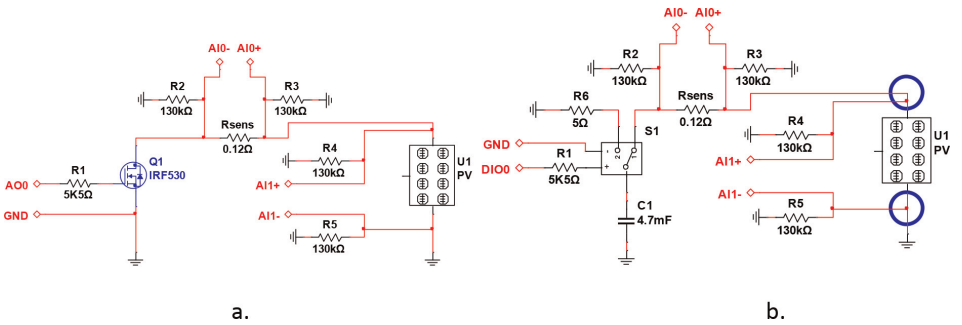


Figure 20.
The circuits schemata for I-V measurement. a. MOSFET method, b. capacitor method.

output channel used for generating a ramp voltage signal applied to the MOSFET gate, ***AI0-*** – ***AI0+*** and ***AI1-*** – ***AI1+*** are the first and second analog input channels configured in differential connection modes and are used for measuring the current and voltage generated by the PV panel. ***DIO0*** is the first digital line of the board configured as the output and is used to control the ***S1*** switch.

Based on the MOSFET circuit schemata, a PCB was designed and build as shown in **Figure 21**, and the entire system for a PV panel characterization is shown in **Figure 22**. The used DAQ board is NI USB 6215.

A very simple software application was developed based on the DAQmx library (**Figure 23**). The block diagram of the application is shown in **Figure 23a** and contains four zones. The first zone, (1), is dedicated to the analog signal generation used for MOSFET gate control. It contains the standard line of DAQmx subVIs for finite samples generation. The second zone, (2), is dedicated to the analog input measurement used for obtaining the current and voltage generated by the PV panel. It contains the standard line of DAQmx subVIs for finite samples acquisition on multiple channels. The third zone, (3), is used for building the signal that controls the MOSFET gate. A *Triangle Waveform.vi* was used and configured for a quarter or half period

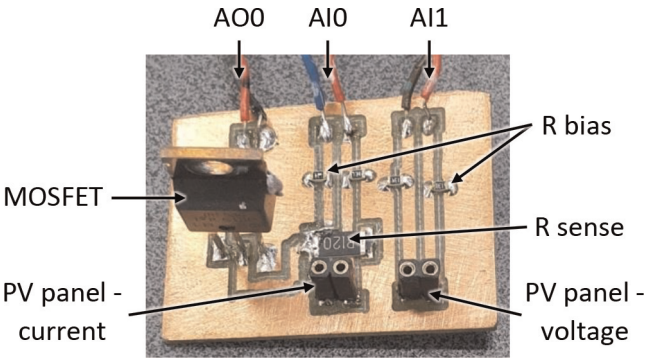


Figure 21.
The electronic circuit based on MOSFET.

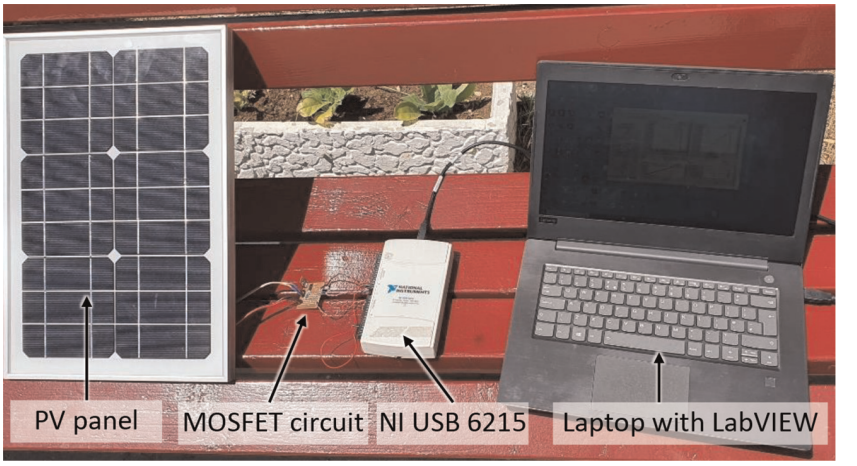
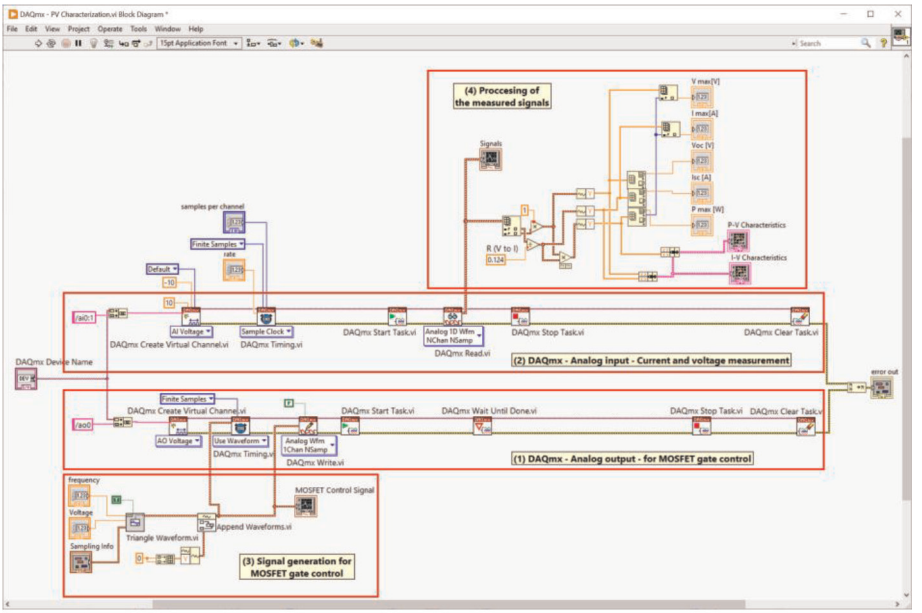
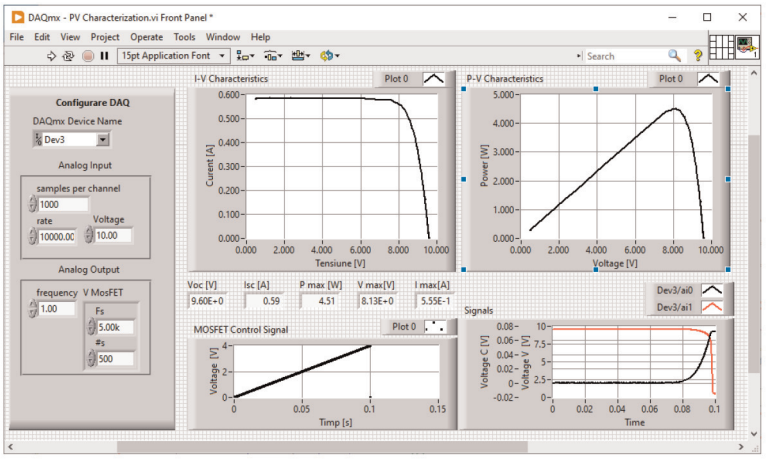


Figure 22.
The system for I-V characteristic measurement of a PV panel.



a.



b.

Figure 23.
The application for PV panel characterization based on the MOSFET method.

generation, through the values of the *frequency* and *sampling info* controls. *Append Wafeform.vi* was used to add the 0 value at the end of the generated signal in order to put the MOSFET in the blocked state so that the PV panel could be placed in the open circuit state at the end of the application execution. The fourth zone, (4), processes the measured signals and extracts the important parameters of the PV panel, such as: $\boldsymbol{P_{max}}$, $\boldsymbol{I_{max}}$, $\boldsymbol{V_{max}}$, $\boldsymbol{I_{SC}}$, and $\boldsymbol{V_{OC}}$. The ***R (V to I)*** constant is used for the conversion of the measured voltage across $\boldsymbol{R_{sense}}$ to the current. By multiplying the measured PV panel voltage and the determined current, the generated power is calculated. The I-V and P-V characteristics are determined and displayed in two graphs.

The same software application and load can be used to measure the I-V characteristics of the TEG in the case that a TEG replaces the PV panel and a temperature difference is ensured between its sides.

In [9], the implementation of the capacitor method is discussed. For better results, a more complex circuit was developed using an electromagnet relay for connecting the capacitor with the PV panel and a self-defined conditioning system based on AD8221 instrumentation amplifiers. For testing, a NI ELVIS II platform was used; see **Figure 24**. The LabVIEW application allows the comparison between data obtained through real measurements and the Multisim simulation directly in LabVIEW. On the measurement section, the actual application replaces zones (1) and (3) with one that controls the DIO channel, ***DO0***, which allows the activation of the electromechanical relay (**Figure 25**). The second zone remains almost the same with the difference of introducing the *DAQmx Start Trigger.vi,* which sets the start of measurements on an analog input channel only when the DO channel sends the relay activation command. Being a polymorphic VI, it was configured on the *Start Analog Edge* option from its dropdown menu. The VI inputs are set on the first channel as a source, rising slope, and 0.05 as level. When the relay is activated, the capacitor being fully discharged, it starts charging, and therefore, the current through the circuit passes from zero to short circuit value, which triggers the measurements. Using a *Case structure*, the application can be used with or without trigger. The application has an event-based architecture implemented with the help of an *Event structure* and a *While loop*. The defined event cases are:

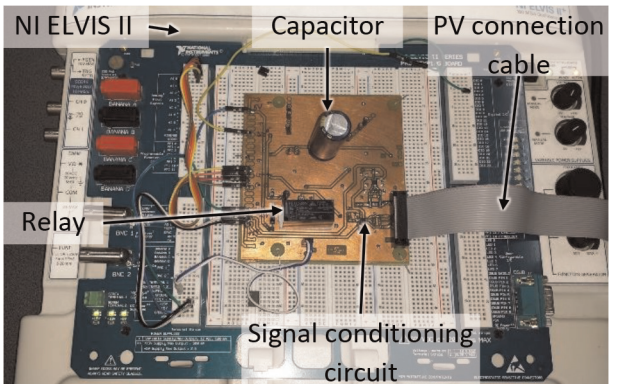


Figure 24.
The circuit for the capacitor method.

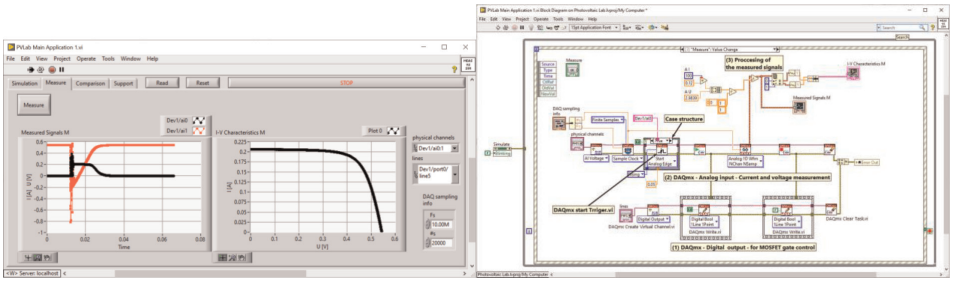


Figure 25.
The LabVIEW application for the capacitor method implementation.

1. ***Measure*** – allows measuring the I-V characteristics of a real PV panel;

2. ***Simulate*** – allows obtaining the I-V characteristics of a simulated PV panel;

3. ***Analyze*** – dedicated to comparing the real and simulated data;

4. ***Read*** – allows importing data from a previous saved file;

5. ***Reset*** – used for clearing all data for a new experiment;

6. ***Stop*** – clears all tasks and stops the application.

John Paliotta, "Software Quality and the Industrial Internet of Things: Why It Matters NOW," Embedded System Engineering, July 6, 2015. The RELab board dedicated for characterization of three RES is described in the paper "Design and implementation of RELab system to study the solar and wind energy" published in the *Measurement* journal [10]. The developed board is a modular one and can be used with three different devices as follows: NI ELVIS II, NI myDAQ, and Ni myRIO (**Figure 26**). By changing the board modules, the purpose of the RELab board can be configured for studying PV cells, small wind turbines, or small solar thermal panels. The software application was developed as an open LabVIEW project including some VIs as laboratory applications grouped on the three RES. The platform allows performing more than 30 lab experiments, out of which 21 are for PV cells, 6 for wind turbines, and 5 for solar collectors. An example of a lab experiment application is shown in **Figure 27**. The main architecture of the lab application is an event-based one. The user interaction with the user interface decides which state is executed. For example, by pressing the buttons ***Start***, ***Open,*** or ***Save***, the application will execute the corresponding state, but also a specific state will be executed if the cursor of the ***I-V Characteristic*** graph is used. At the beginning, the application has a section that starts a web browser in the user interface, loading the lab work web page. In this manner, the user has access to the theoretical aspects of the lab and also the steps to follow to perform the laboratory work. In the measurement state called *Start,* there are three lines dedicated to reading the data from the PV cell and sensors. The first line reads the current and voltage generated by the PV cell and also its temperature based

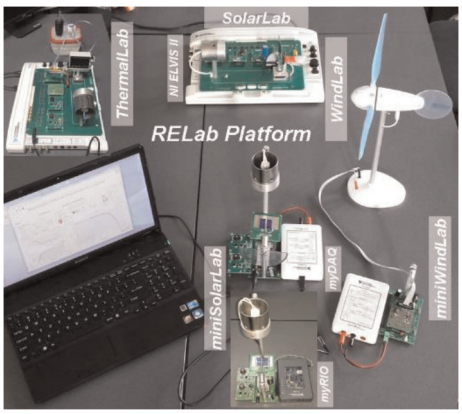


Figure 26.
The RELab platform.

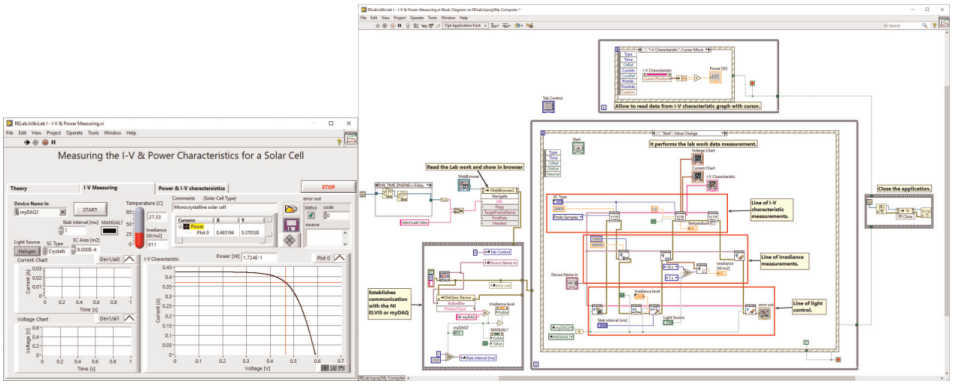


Figure 27.
A lab experiment application for RELab board.

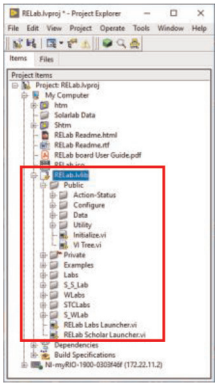


Figure 28.
A lab experiment application for RELab board.

on analog inputs. The second line has the purpose of reading the data from the irradiance sensor. This line uses an analog input of the NI ELVIS platform and a counter input for NI myDAQ and NI myRIO. The third line has the goal of controlling the level of irradiance, done based on varying the voltage level generated on an analog channel.

For an easier programming, the LabVIEW project was developed on a driver structure as can be seen in **Figure 28**, highlighted with a red frame. Based on the included subVIs, the user can develop new lab work applications based on their own concepts and methods.

3.2 LabVIEW and NI cRIO

For complex applications, a characterization system was developed along the NI cRIO 9074 platform. The system has four independent channels so that it allows the characterization of four synchronous RES. A self-developed electronic load with four independent channels was developed based on the capacitor method. NI cRIO is an industrial hardware dedicated for measurements, monitoring, and industrial control.

The NI cRIO 9074 has the following characteristics: 400 MHz CPU, 128 MB DRAM, 256 MB Storage, and 2 M Gate FPGA. The platform can act as an embedded system running applications for control and monitor without a PC connection.

The LabVIEW project is shown in **Figure 29a**. It can be seen that there are three levels of the project:

1. My computer branch (PC level) – which contains applications that are running on the PC. The *PC Main v1.vi* has the goal of showing and sending the necessary parameters to the embedded system. Also, the application accordingly processes the data received for I-V characteristics, extracting the main parameters as V_{OC}, I_{SC}, P_{max}, V_{max}, and I_{max} for all four channels.

2. NI-cRIO9074 branch – Real Time (RT) level – dedicated to running the application on the RT processor of the NI cRIO platform. The *RT Main V1.vi* takes data and sends data to the FPGA application and also stores the data on the internal permanent memory under TDMS format.

3. FPGA Target branch – FPGA level – contains the applications that are running at the FPGA level. The *FPGA Main v1.vi* has the goal of data acquisition from the analog input channels (current, voltage, and temperatures) and generates digital states for enabling/disabling the electronic load channels.

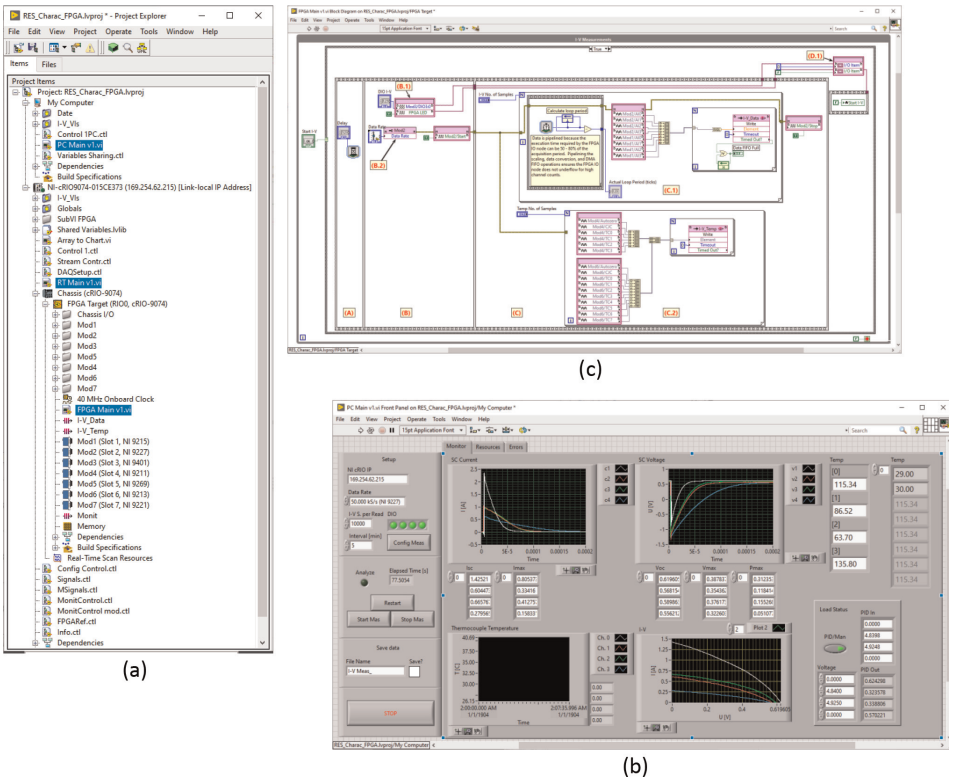


Figure 29.
NI cRIO application, a) LabVIEW project; b) PC user interface; and c) the block diagram of the FPGA application.

The block diagram of the FPGA application is shown in **Figure 29c**. The application runs continuously using an infinite while loop containing a case structure that allows controlling the start of the measurement – which happens if the **Start I-V** variable receives the True value from RT application.

The *Sequence* structure allows controlling the steps of measurements. In the first frame (A), a settable delay is introduced to make sure that the RT time application passes to the monitoring state for receiving data. Frame B sets the used channels and also activates the user LED of the cRIO platform for knowing the state of the application, with the debugging purposes (B.1). Also, in this frame, the acquisition rate and the starting of the acquisition are settled (B.2). In Frame C, the data acquisition is done for current and voltage using the loop C.1 and for temperature, the loop C.2. The acquired measurements are transferred to the RT application using two FIFO queues: **I-V_Data** and **I-V_Temp** (see **Figure 29a**). The transfer is based on the DMA technique. When the required number of data is achieved, the data acquisition is stopped, and the used channels and the user LED are disabled (D.1).

The code of the RT level application is shown in **Figure 30**. The code is based on two while loops: first ensures the communication between this application and the application running on the PC (**PC-RT Loop**), and the second is dedicated to controlling the entire system (**RT-FPGA Loop**). This loop implements a state-machine architecture, which has the following states: **Idle**, **Init**, **Config**, **Monit**, **Status**, **I-V**, **Data Save**, **Stop,** and **Restart**. The **Init** state deploys the bitfile (A) obtained by compiling the FPGA application. At the same time, a TDMS file with a predefined name is opened (B). By reading the configuration shared variables in the **PC-RT Loop**, the running parameters of the RT and FPGA application are fixed in the **Config** state. The I-V characteristic is measured at a predefined interval of time in the **I-V** state (**Figure 31**). The line (A) is dedicated to the starting DMA transfer using the

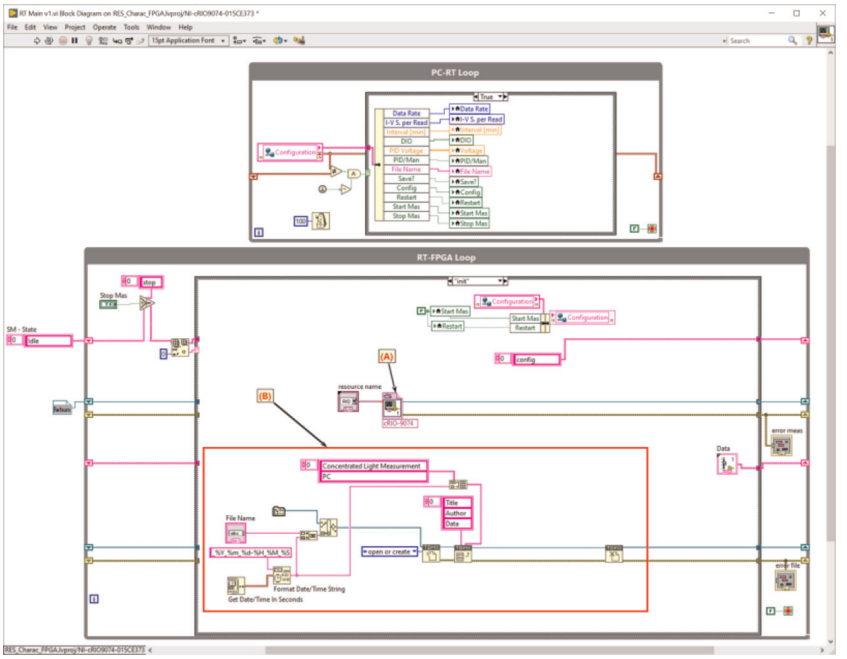


Figure 30.
The block diagram of the RT application.

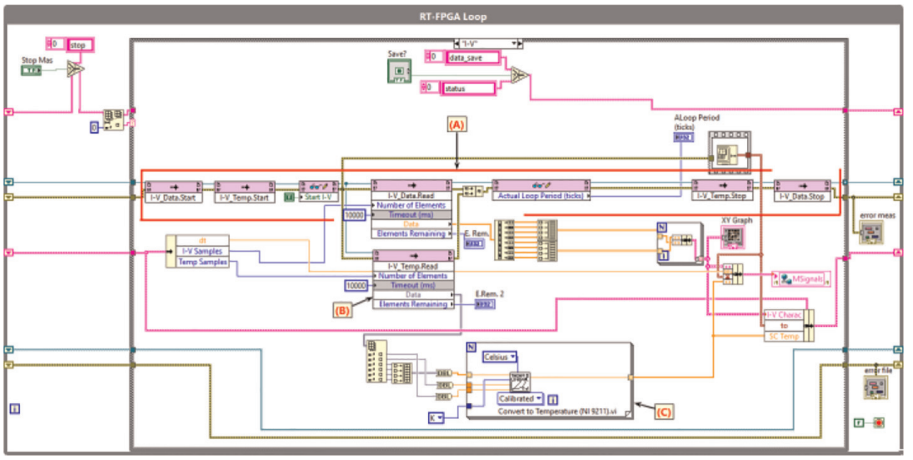


Figure 31.
The I-V state of the RT application.

FIFO queues. By sending the true value to the **Start I-V** button on the FPGA application, the data acquisition is started; then, through the *I-V_Data Read* (A – line) and *I-V_Temp Read* (B – line) nodes, the measured data are transferred from the FPGA to the RT application. Within the (C) loop, the read voltages from the K-type thermocouples are converted to temperature, in the Celsius scale. The read data are transferred to the PC application through the shared variables Measured Signals (**MSignals**) and through **Data** cluster to the **Data Save** state for storing the data in the opened TDMS file.

The shared variables are programmatically called in the PC application (see **Figure 32**) using the appropriate path and the Shared Variables subVIs (*Functions*> > *Data Communication* > *Shared Variable*): *Open Variable Connection*, *Read Variable*, *Write Variable,* and *Close Variable Connection* (**Figure 33**).

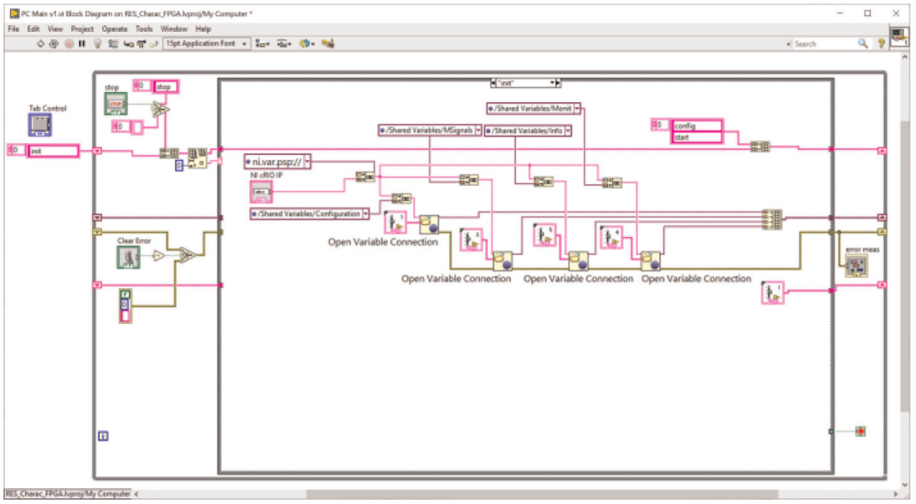


Figure 32.
*The **Init** state of the PC application.*

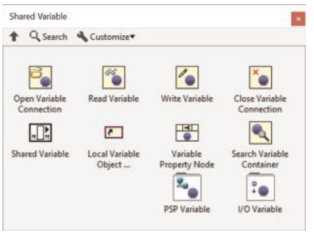


Figure 33.
The shared variable palette.

After the application is started, even if the PC is disconnected, the NI cRIO platform continues to measure and store data based on the defined parameters. When the PC is reconnected, the application continues to run without any interruption.

The saved files on the NI cRIO platform can be downloaded using the FTP protocol using File Explorer or any other FTP client.

4. Monitoring and industrial application

This section describes a real-life application, where a dedicated monitoring system is deployed on a PV-TEG to understand its benefits when supplying a household, from both the energy savings (avoiding producing energy from conventional sources) and the power quality perspectives. Once data is collected and processed, the next step is to publish data by means of the IIoT.

IIoT develops the central nervous system of a smarter world, building on equipment and devices that can measure and interact with humans [11]. According to the IEEE Standards Association, IIoT will become one of the drivers of growth in a wide range of technologies, with huge business potential, valued at $ 14.2 trillion in 2030. Within IIoT, data communication follows three distinct paths [12]: machine-machine (M2M), man-machine (H2M), and machine-smartphone (M2S), where the smartphone can be any other device with a touch interface, for example, a tablet.

4.1 System's structure

The deployed PV-TEG system contains 16 monocrystalline JA Solar 340 W panels, connected to a Fronius Primo 5.0 power inverter, generating up to 5.5 kW, as seen in **Figure 34**. This is the so-called *prosumer* configuration, where no storage unit is installed, so the energy excess is delivered to the power utility provider.



Figure 34.
Deployed PV panels on the household's roof, Google maps and photo.

The implemented measurement application has the main objective of gathering and exposing the process data according to the IIoT concept, which ensures visibility to various users. Using cloud technologies, the system achieves higher performance rates in terms of reliability and security. An alternative which has been considered would have used an on-premises solution, but the latter one raises additional costs and has a higher degree of sophistication, with a lower efficiency in operation.

Regarding the measurement hardware, the most important criterion considered in this project is its performance, that is, processing power, memory, high-quality DAQ modules, programming capabilities, communication interfaces, and internet integration.

The chosen solution is based on the NI Compact RIO platform, being deployed on two systems using NI cRIO 9033 controllers, which support NI Linux RT. The cRIOs 9033 are equipped with voltage (400 V RMS) and current (50 A RMS) modules, among other dedicated ones, as seen in **Figure 35**.

The investigated system has several operating requirements, influencing its design: (i) 24/7 operation; (ii) autonomous operation; (iii) easy to program and maintain; (iv) easy to expand without significant impact on the measurement system's performance; and (v) data access: web based, secured.

The application is multilayer type, distributed, according to the methods represented in **Figures 36** and **37**, respectively. NI LabVIEW™ is the software development environment used throughout the entire software implementation:

- Primary level of data acquisition, DAQ. The implementation is at the FPGA level, facilitating the collection of data at higher speeds, so that the data analysis support is provided at the upper, RT level

- Level of analysis of process data: (i) aggregation of data collected at the FPGA level, high-performance communication routines with the FPGA infrastructure; (ii) synchronization of collected data; (iii) effective data analysis: total harmonic distortion (THD), phase shift; and (iv) calculation of process quantities: effective values, frequency, powers, phasors, etc.

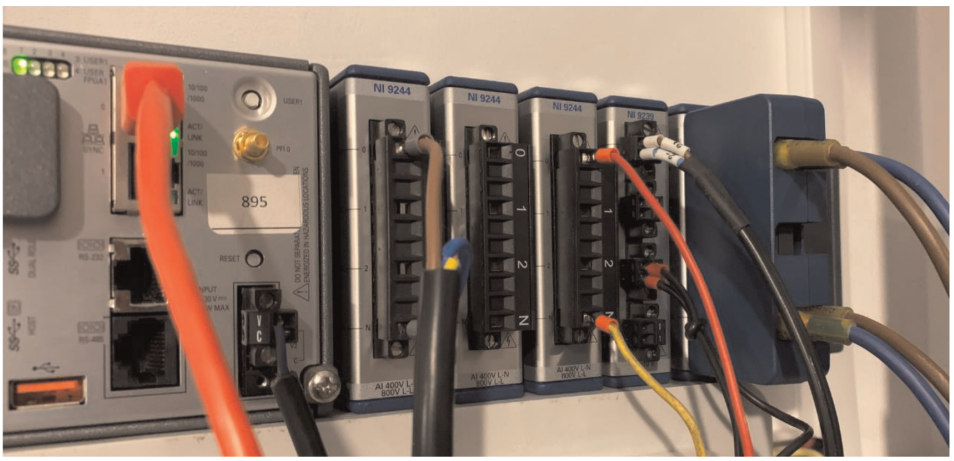


Figure 35.
NI compact RIO in use, monitoring the PV-TEG.

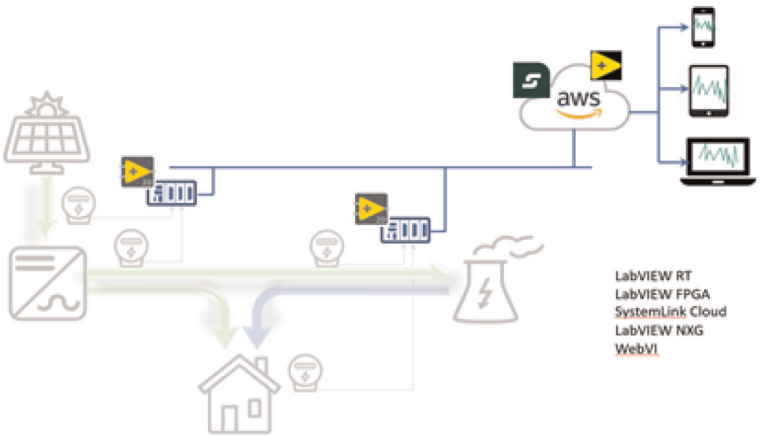


Figure 36.
PV-TEG with its associated monitoring system.

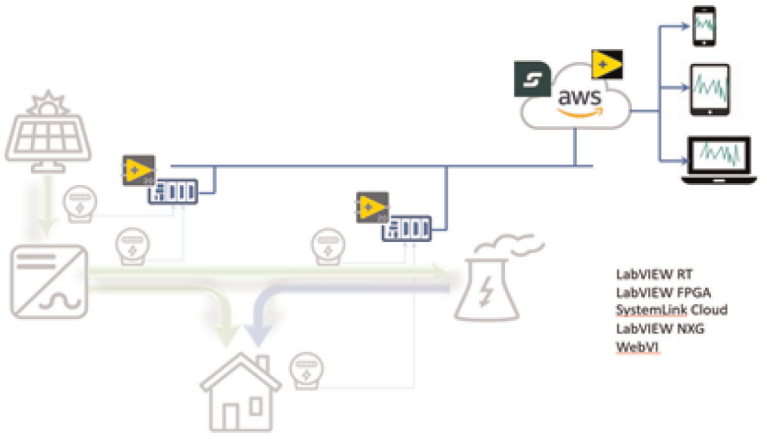


Fig. 36 PV-TEG with its associated monitoring system

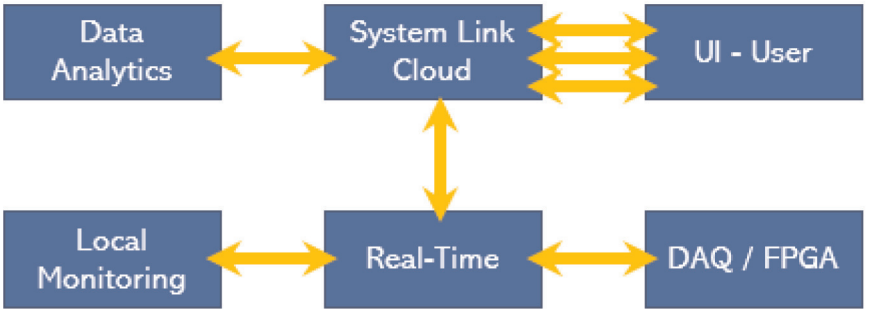


Figure 37.
Multilayer SW architecture of the proposed system.

- Local display of data for monitoring and maintenance
- Primary redundancy level implementation
- Additional analysis support
- Secured communication with the cloud application
- Communication with field equipment using the OPC UA protocol, mainly system services

At the top level, the application contains several modules:

- Data server module: definition, configuration, and operation of tags (process data)
- Cyber security module: defining the keys and security policies that address the communication with the primary measurement equipment
- User interface module: User-exposed application development using high-performance toolsets
- Data analysis module: stand-alone, workstation-level implementation: cloud interface, advanced analysis, post-data processing, statistical and process optimization, and cloud data exposure to facilitate third-party access.

4.2 Data acquisition software

The reconfigurable FGPA section is in charge of the high-speed data readings from sensors, passing these data along to the next level, the Real Time part of the application, by using dedicated FIFO mechanisms, as seen in **Figure 38**.

The Real Time part, the connected instance to the primary DAQ-FPGA loop, manages the lower speed part of the application, dealing with: (i) data retrieval from the DAQ FIFO, served by the FPGA app; (ii) data analysis related to the electrical specific values: power calculation – active, reactive, power factor, total harmonic distortion, efficiency, etc. Extra analysis functions could be easily added, with almost no penalty to the computing performances. Data retrieving section could be seen in **Figure 39**. Data collected from the FPGA primary source is copied to a shared variable structure, which handles the access of the other sections of the software.

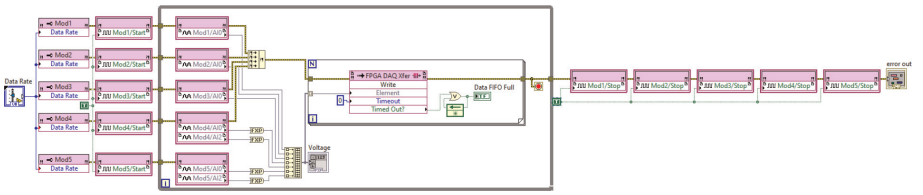


Figure 38.
DAQ implementation, FPGA level.

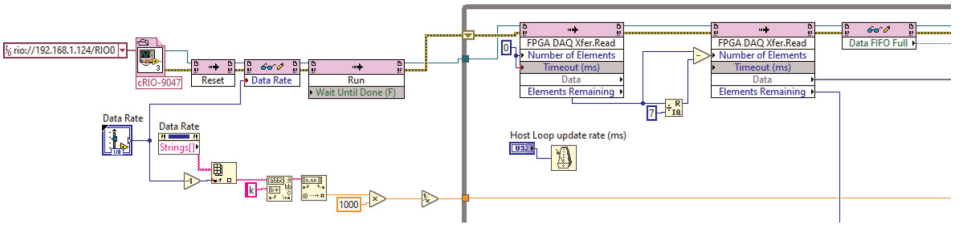


Figure 39.
DAQ implementation, RT level.

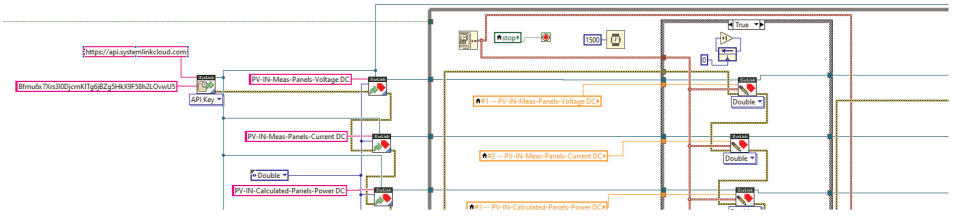


Figure 40.
Cloud communication protocol implementation, RT level.

Just to add more details to the previous statement, **Figure 40** depicts the communication implementation toward the AWS cloud using the NI SystemLink Cloud solution.

In this figure, one can see how shared variables, which support the communication protocol between different pieces of the software implementation, are pushed up to the NI SystemLink Cloud implemented using AWS support.

4.3 Cloud-based user interfaces

Figure 41 captures the user interface, which serves the embedded measurement system. One can observe the project window (left side of the left screen), accompanied by the main dashboard screens, which contain process data plots – raw data from system sensors, calculated values (power, power factor, efficiency, and financial data), plus other parameters describing the system's health. All this information is live data, allowing the user to efficiently manage the system, supporting business decisions, and also serving as a maintenance tool.



Figure 41.
DAQ user interfaces, system monitoring, data analysis and maintenance.

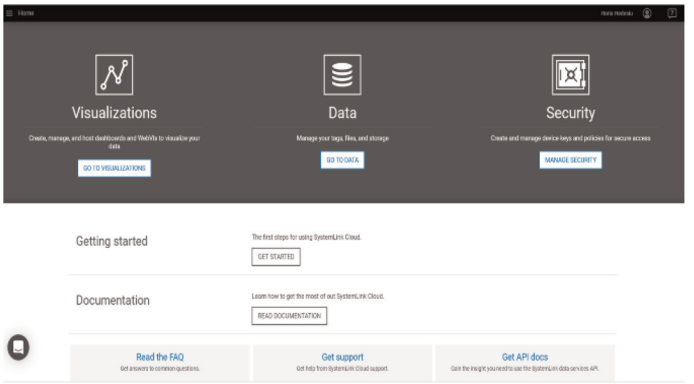


Figure 42.
NI SystemLink cloud landing page.

When designing the monitoring application, the final part of the software development has to manage the process data received from the field equipment. PV-TEG process data pushed to the cloud is organized in so called 'tags.'

The NI SystemLink Cloud server benefits of several built-in capabilities, see **Figure 42**: Data *Management* (tags – process information), *Cybersecurity* – API keys and policies, *User Interfaces* – various templates with more or less displaying capabilities, or user-defined interfaces developed using the state-of-the-art WebVIs created using LabVIEW NXG.

The LabVIEW NXG UIs are a more refined way to display process data coming from field. An important advantage is the capability of processing information through the LabVIEW diagram, whereas the other options are display only. As seen in **Figure 43**, data retrieved from NI SystemLink Cloud is processed (power-related math, financial info, THD, etc.) and displayed, either as instantaneous values or as waveforms.

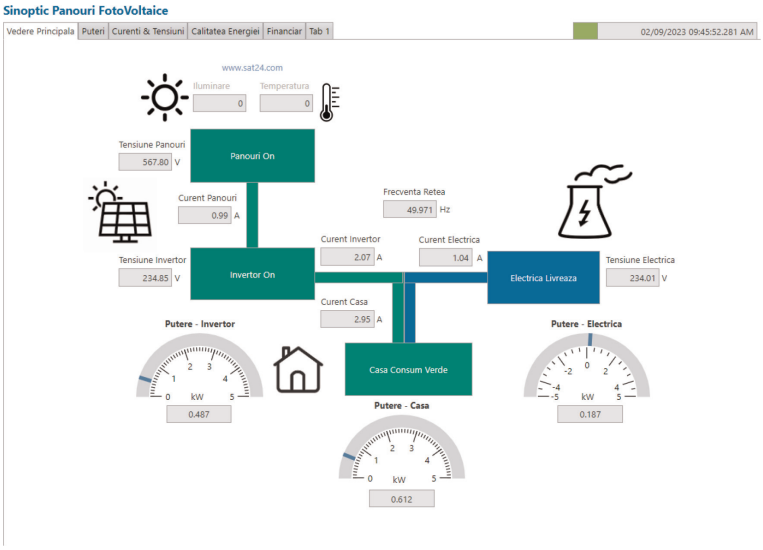


Figure 43.
Cloud-based UI, LabVIEW NXG diagram.

Such examples are shown in **Figures 44-46**, where one can contemplate a general view of the PV-TEG-based power generation system, containing instantaneous information describing the system's behavior (voltages, currents, powers, etc.), followed

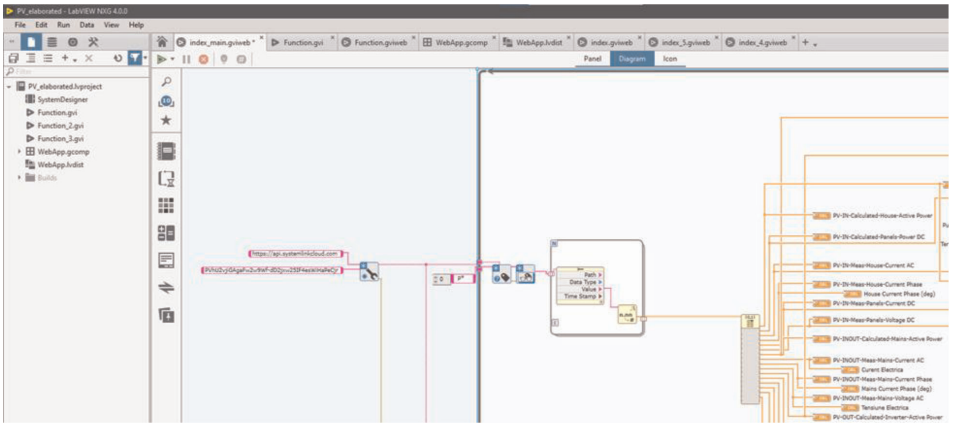


Figure 44.
Cloud based UI, LabVIEW NXG diagram.

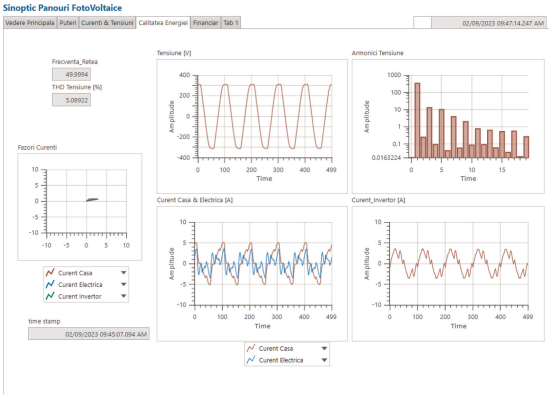


Figure 45.
Cloud based-UI, electrical power quality tab view.

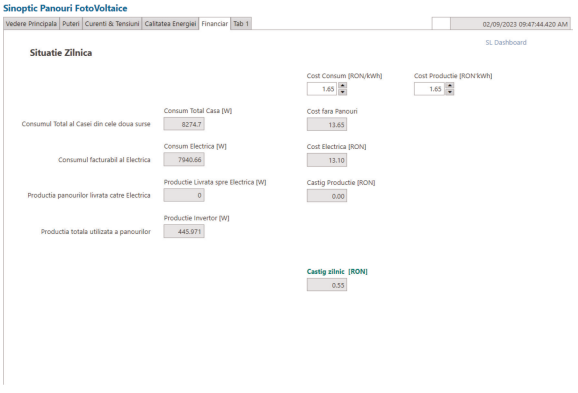


Figure 46.
Cloud-based UI, financial daily report tab view.

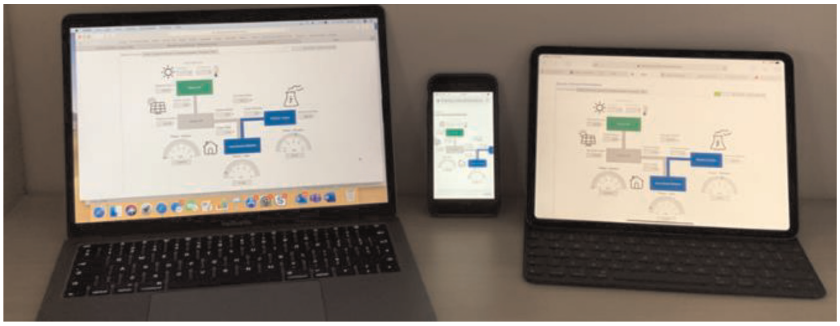


Figure 47.
IIoT NI SystemLink cloud-based interfaces, sys monitoring, data analysis, running on portable and mobile devices.

by a series of other screens where different trends are displayed, depicting the dynamics of the monitored system. All these parameters are retrieved from the cloud DB NI SystemLink Cloud, making the whole development process streamlined and effective. As illustrated in **Figures 45** and **46**, real-time data of selected parameters serve as a powerful analysis tool that allows the user to observe waveforms and to perform electrical power quality and financial analysis on raw and processed data.

The NI SystemLink Cloud [13] implementation brings in consistent advantages for data security and access. User Interfaces are tailored in such a manner that fit different devices, starting with computers and ending with portables (smart phones and tables), as shown in **Figure 47**.

The PV-TEG monitoring application is live, the access is granted to all interested users by simply accessing this link: https://hosting.systemlinkcloud.io/webapps/72a70649-5148-485e-b689-04b859fd1cf5/content/ApplicationFiles_64/index.html

Compared with a classical web server implementation, the cloud solution serves as *de facto* IIoT approach, being scalable, easy to maintain, and very efficient in information dissemination to a large spectrum of users, regardless of their technical abilities.

Besides the technical/engineering data, financial information is also available, as per user needs. These numbers play a crucial role, of course, in assessing the overall effectiveness of the PV-TEG system while running.

5. Conclusions

This chapter presents different solutions for implementing different simulations for PV and TEG modules, real characterization of the PV and TEG modules based on real hardware, and a solution for implementing a real monitor system for a photovoltaic home application.

Starting from the fact that the energy demand has increased worldwide and therefore new energy sources or increasing the efficiency of the existing ones is necessary, the RES has become very important nowadays as clean solutions instead of fossil energy sources. Therefore, the study, the understanding, and the usage of the RES are very important, and also it is very important to have solutions for training the personal for implementation and maintenance of the RES. Using LabVIEW along with other tools and hardware devices represents a very versatile solution.

LabVIEW allows different approaches for RES simulation through direct coding, using simulation toolkit or in combination with other software applications like NI Multisim.

Interacting with real RES is very facile in LabVIEW by using the very powerful API for data acquisition and the appropriate data acquisition hardware, from simple data acquisition board, like NI USB 6215 to NI ELVIS platform or more complex NI cRIO.

The real-life application presented in this chapter stands for a rapid prototyping measurement system using embedded equipment instrumented with graphical programming tools, starting from lower-level operation (DAQ), all the way up to its User Interfaces deployed in Cloud. The integrated way of data manipulation, from sensor to monitors and computer displays, allows the users to learn efficiently about the system's behavior and how to improve its performances while using.

All these collected data could serve for future analysis and optimization using more sophisticated techniques, like artificial intelligence, which could easily connect to IIoT-based solutions developed with LabVIEW™.

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